

Unit04: Installation and development environment overview

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Objectives

- Understand the use of Rapid Miner.
- Know the different Data Mining products.
- Learn the steps for installing Rapid Miner.
- Understand what WEKA is.
- Learn the installation process of WEKA.

Introduction

Rapid Miner provides an environment for machine learning and data mining processes. It follows a modular operator concept which allows the design of complex nested operator chains for a huge number of learning problems. Allows for the data handling to be transparent to the operators.

Weka is an open-source tool designed and developed by the scientists/researchers at the University of Waikato, New Zealand. WEKA stands for Waikato Environment for Knowledge Analysis. It is developed by the international scientific community and distributed under the free GNU GPL license. It provides a lot of tools for data preprocessing, classification, clustering, regression analysis, association rule creation, feature extraction, and data visualization. It is a powerful tool that supports the development of new algorithms in machine learning.

4.1 RapidMiner

The idea behind the Rapid Mining tool is to create one place for everything. RapidMiner is an integrated enterprise artificial intelligence framework that offers AI solutions to positively impact businesses. It is used as a data science software platform for data extraction, data mining, deep learning, machine learning, and predictive analytics. It is widely used in many business and commercial applications as well as in various other fields such as research, training, education, rapid prototyping, and application development. All major machine learning processes such as data preparation, model validation, results from visualization, and optimization can be carried out by using RapidMiner.

Facilities of RapidMiner

1. Rapid Miner provides its collection of datasets but it also provides options to set up a database in the cloud for storing large amounts of data. You can store and load the data from Hadoop, Cloud, RDBMS, NoSQL, etc. Apart from this, you can load your CSV data very easily and start using it as well.

2. The standard implementation of procedures like data cleaning, visualization, pre-processing can be done with drag and drop options without having to write even a single line of code.
3. Rapid Miner provides a wide range of machine learning algorithms in classification, clustering, and regression as well. You can also train optimal deep learning algorithms like Gradient Boost, XGBoost, etc. Not only this, but the tool also provides the ability to perform pruning and tuning.
4. Finally, to bind everything together, you can easily deploy your machine learning models to the web or mobiles through this platform. You just need to create user interfaces to collect real-time data and run it on the trained model to serve a task.

RapidMiner Products

The following are the various products of RapidMiner:

- RapidMiner Studio
- RapidMiner Auto Model
- RapidMiner Auto Model

RapidMiner Studio

- With RapidMiner Studio, one can access, load, and analyze both traditional structured data and unstructured data like text, images, and media.
- It can also extract information from these types of data and transform unstructured data into structured.

RapidMiner Auto Model

- Auto Model is an advanced version of RapidMiner Studio that increments the process of building and validating data models.
- Majorly three kinds of problems can be resolved with Auto Model namely prediction, clustering, and outliers.

RapidMiner Turbo Prep

- Data preparation is time-consuming and RapidMiner Turbo Prep is designed to make the preparation of data much easier.
- It provides a user interface where your data is always visible front and center, where you can make changes step-by-step and instantly see the results, with a wide range of supporting functions to prepare the data for model-building or presentation.

4.2 Downloading of Installation of RapidMiner

RapidMiner provides a data science platform to help you drive real business impact. Choose the solution based on your preferences.

Where to get it? To install RapidMiner Go to:

<https://rapidminer.com/get-started/>

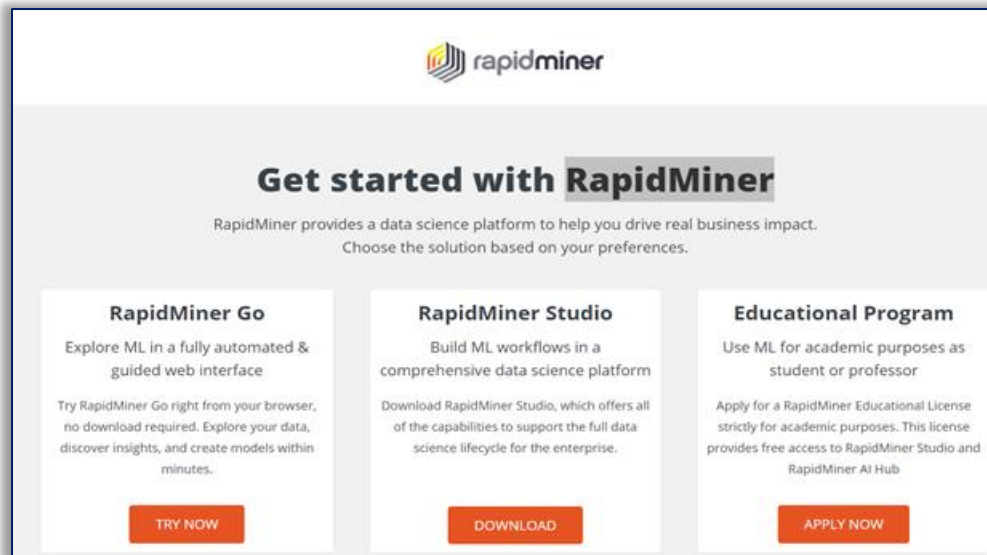


Figure 1: Get Started with Rapid Miner

RapidMiner Installation

And proceed to click on the version of RapidMiner Studio that fits your system:

- Windows 32bits
- Windows 64bits
- Mac OS 10.8+
- Linux

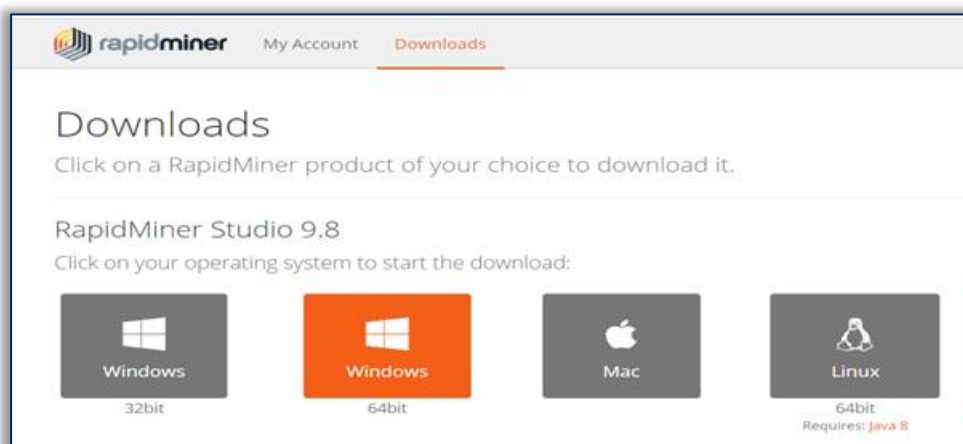


Figure 2: Operating System Selection

Installing RapidMiner Studio on Windows

Follow these simple instructions to run the launcher that installs RapidMiner Studio on Windows.

- Double-click the downloaded file (for example, rapidminer-studio-<version>-win64-install.exe).
- If prompted, allow the program to make changes to your computer. The RapidMiner Studio Setup Wizard appears. Click Next to continue.
- Read the terms of the license agreement and click I Agree to continue.

- Select a destination folder (or leave the default). Please ensure that the folder path does not contain + or % characters. By clicking Install, the wizard extracts and installs RapidMiner Studio. When the installation completes, click Next and then Finish to close the wizard and start RapidMiner Studio.
- Read the terms of the license agreement and click I Accept to continue.

The download will start automatically. When it finishes, click “RUN” so that the installation can begin.

Agree with Terms and Select Destination

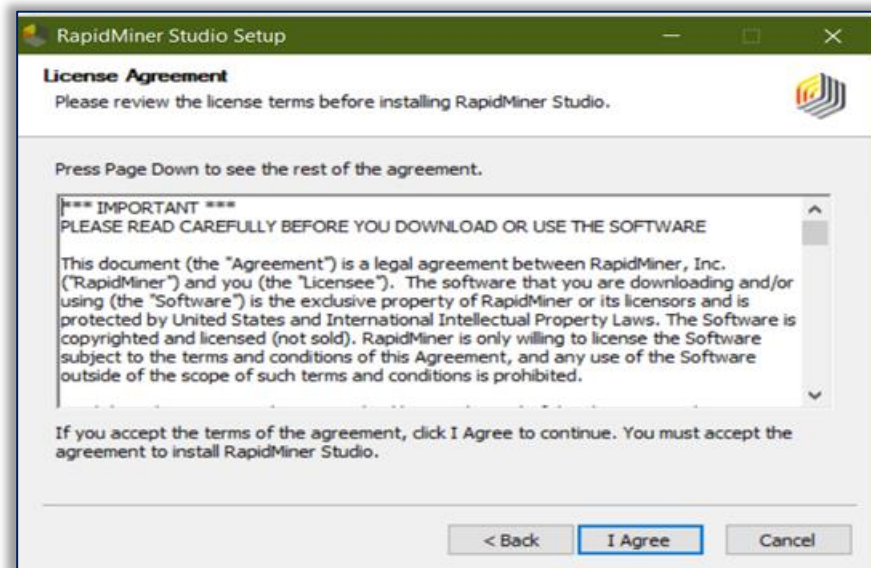


Figure 3: License Agreement

You'll see the welcome screen for the installation. Click “Next” to move to the terms of Use, and if you agree with them, click “I Agree”. Finally, select the destination folder for the application (You'll need 224.5MB of free space to install) and click “Install”.

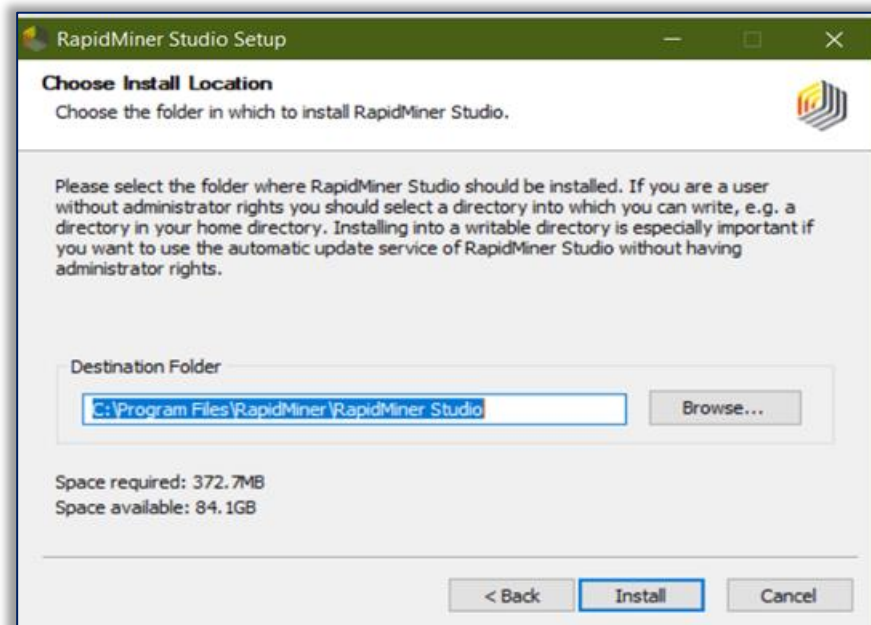


Figure 4: Choose install Location

A progress bar will show the progress of the installation, and when it finishes (takes less than 5 min) you'll see the "Completing the RapidMiner Studio Setup Wizard". Click the Finish button to finish.

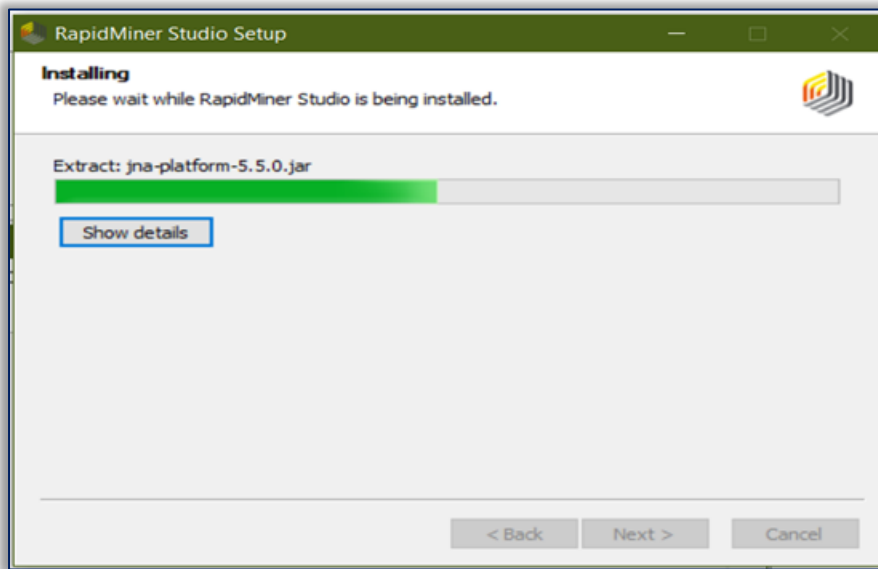


Figure 5: Installation Progress Bar

Click the Finish button to finish the installation process, and congratulations! You are ready to use the application! To open it, just look for it on the desktop, or search for "RapidMiner Studio" on the Windows Start Menu.

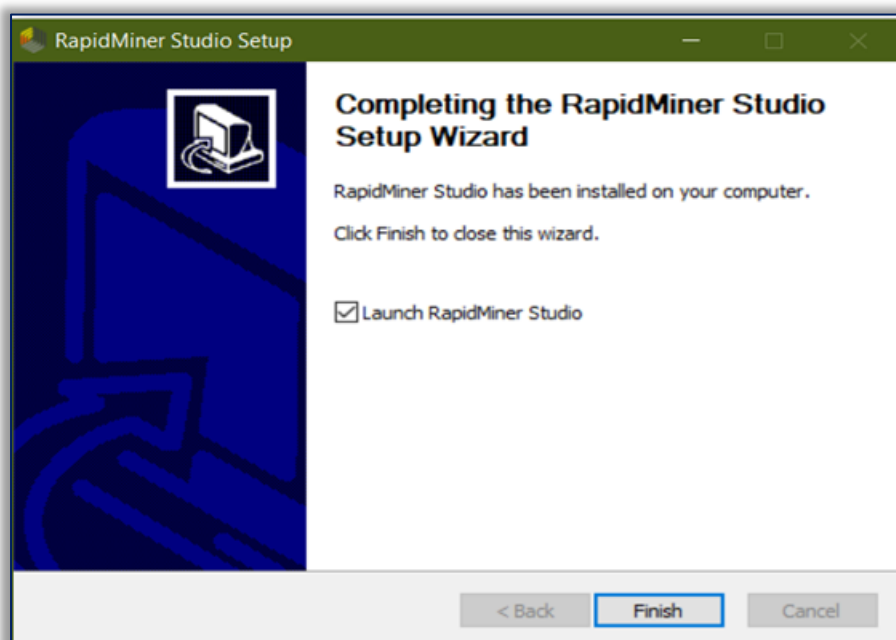


Figure 6: Set-Up Completion Wizard

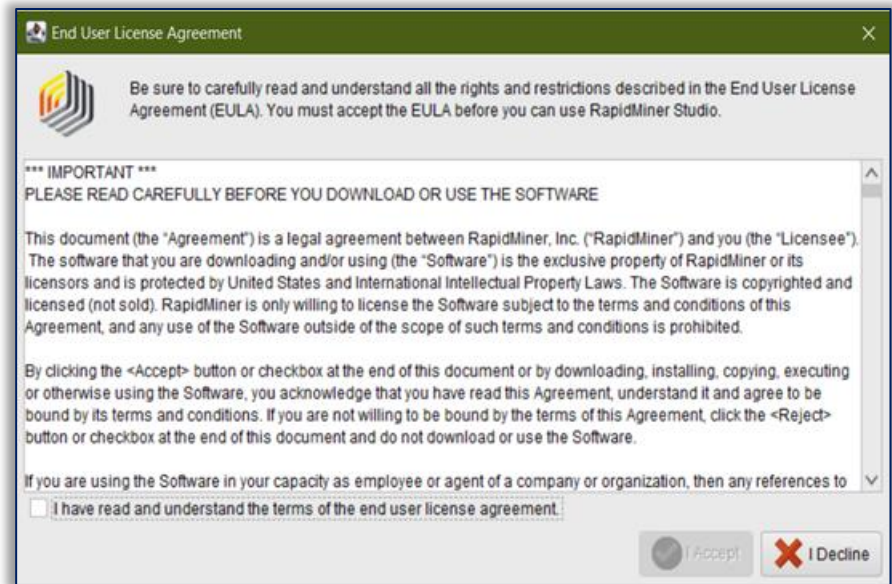


Figure 7: End-User License Agreement



Download and install RapidMiner on your system

A screenshot of the 'Create a RapidMiner account' form. The form has a title bar with the RapidMiner logo and the text 'Create a RapidMiner account'. On the left, it says 'You'll use your RapidMiner Account to access:' followed by a list of benefits: 'the Community forum', 'the Extensions Marketplace', 'free cloud storage', 'product news and updates', and 'product license information'. On the right, there are input fields for 'Email', 'Password', and 'Confirm Password'. Below these fields is a checkbox labeled 'Remember my password' which is checked. At the bottom right, there is a large green button with a checkmark and the text 'Create my Account!'. Below this button is a link that says 'I already have an account or license key'.

Figure 8: User account creation Interface



Getting started with RapidMiner Studio

Once you launch RapidMiner Studio, a Welcome screen appears, prompting with two options:

Table:

Option	Description
Create my Account	Creates an account with the given useful information.
I already have an account or license key	If you previously registered, enter your rapidminer.com credentials to log in so that you can download or install your license(s).

Create RapidMiner account

Creating an account requires that RapidMiner Studio can access the Internet. If you do not have an Internet connection for the application (for example, if prevented by a firewall), create an account through your browser and enter your license key manually.

Complete the steps to create an account by entering your email address and password. Note that the password must be a minimum of six characters.

Immediately, the system sends an activation email to the address that you registered. (Allow email from **RapidMiner** if you do not see an email titled "Verify Your Email" in your inbox.) Periodic checks verify whether your email has been validated.

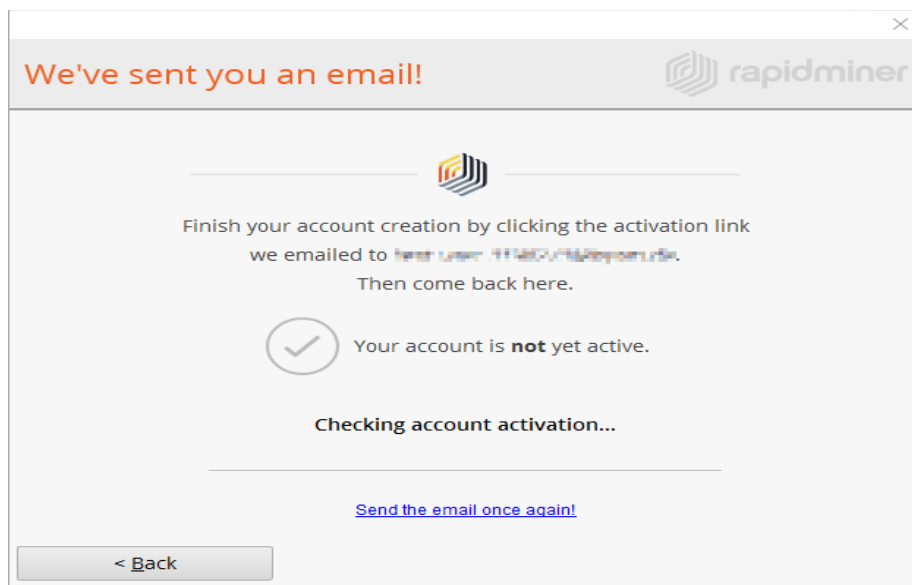


Figure 9:

Once you validate, you'll receive a success message in your browser:

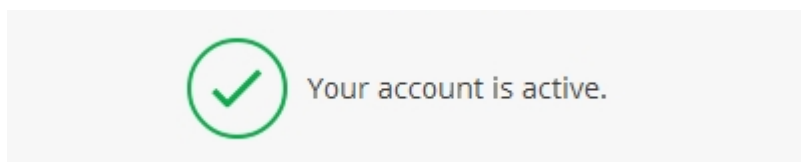


Figure10:

Return to the RapidMiner Studio application. Complete the installation by clicking I'm ready.

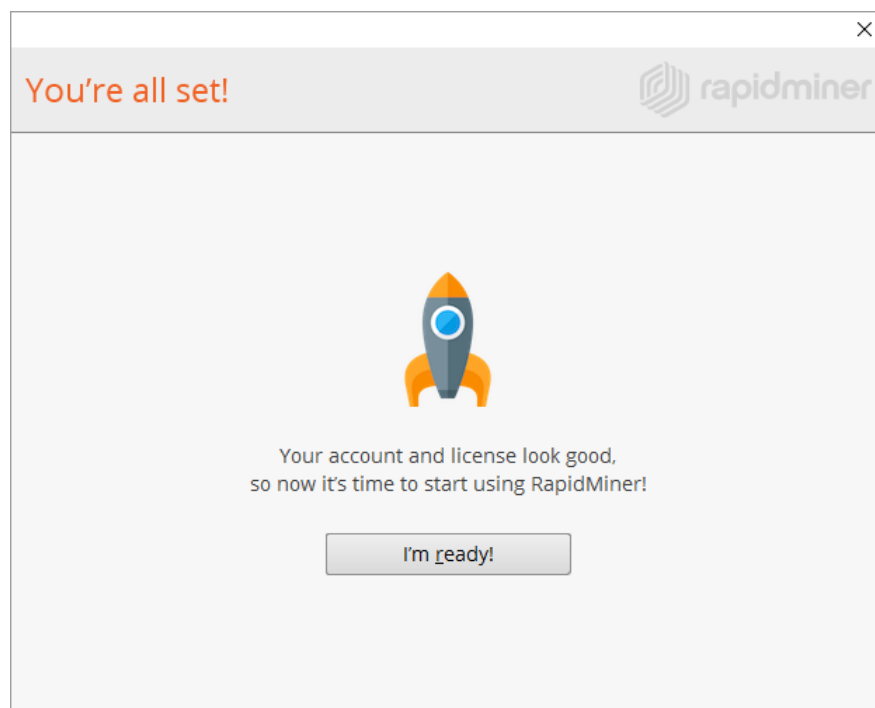


Figure 11

RapidMiner Studio login

If you have already created a RapidMiner account, clicking **I already have an account or license** on the Welcome screen brings up the login screen:

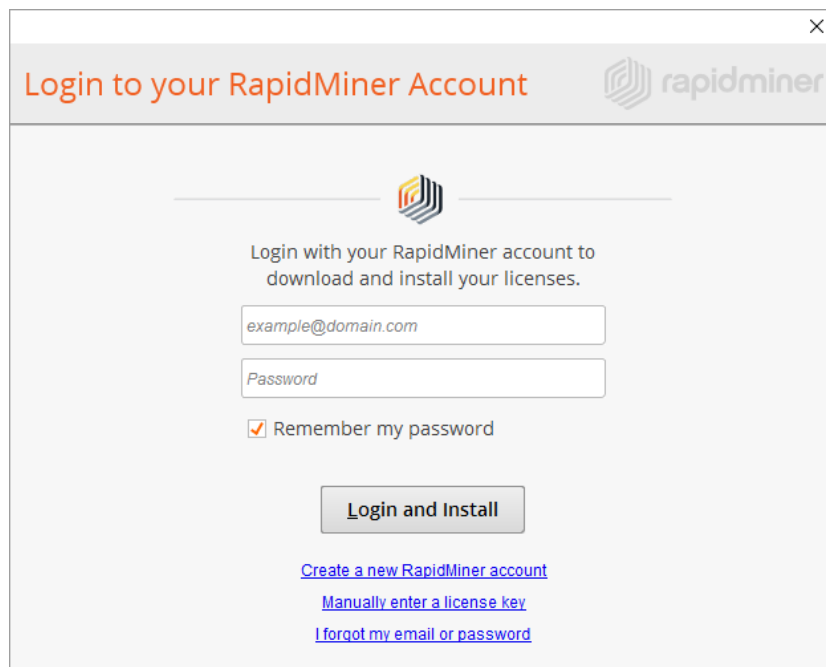


Figure 12: Login Screen

Enter your email address and password to log in with your RapidMiner.com account and then click **Login and Install**.



Create and verify your RapidMiner Account

4.3 Weka

WEKA - open-source software that provides tools for data preprocessing, implementation of several Machine Learning algorithms, and visualization tools so that you can develop machine learning techniques and apply them to real-world data mining problems. What WEKA offers is summarized in the following diagram:

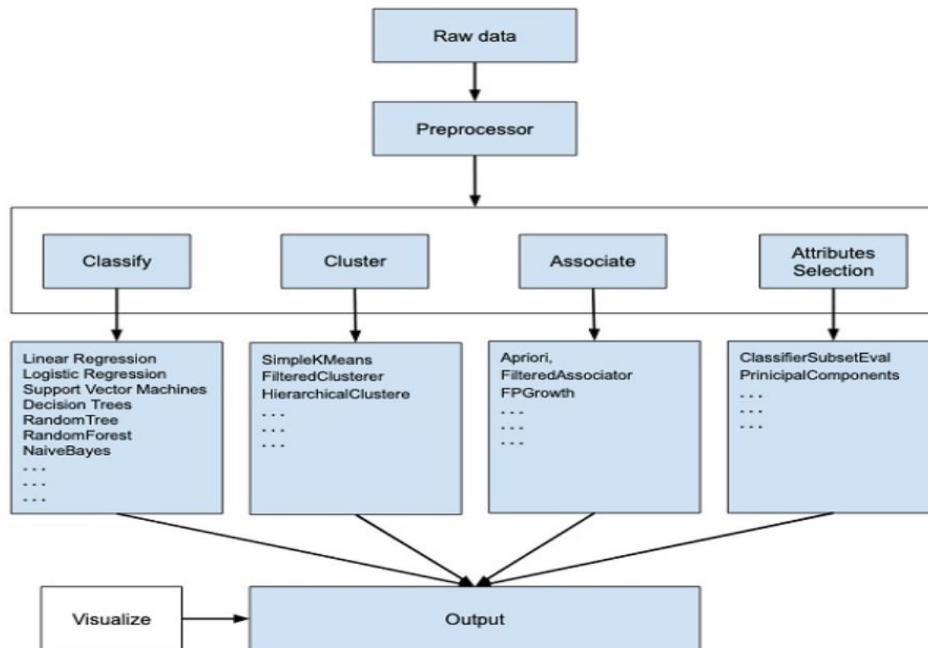


Figure 13: Weka Summary

If you observe the beginning of the flow of the image, you will understand that there are many stages in dealing with Big Data to make it suitable for machine learning –

First, you will start with the raw data collected from the field. This data may contain several null values and irrelevant fields. You use the data preprocessing tools provided in WEKA to cleanse the data. Then, you would save the preprocessed data in your local store for applying ML algorithms. Next, depending on the kind of ML model that you are trying to develop you would select one of the options such as Classify, Cluster, or Associate. The Attributes Selection allows the automatic selection of features to create a reduced dataset. Note that under each category, WEKA provides the implementation of several algorithms. You would select an algorithm of your choice, set the desired parameters, and run it on the dataset. Then, WEKA would give you the statistical output of the model processing. It provides you a visualization tool to inspect the data.

The various models can be applied to the same dataset. You can then compare the outputs of different models and select the best that meets your purpose. Thus, the use of WEKA results in quicker development of machine learning models on the whole.

Weka – Installation

To install WEKA on your machine, visit [WEKA's official website](#) and download the installation file. WEKA supports installation on Windows, Mac OS X, and Linux. You just need to follow the instructions on this page to install WEKA for your OS.

The steps for installing on Mac are as follows –

- Download the Mac installation file.
- Double click on the downloaded **weka-3-8-3-corretto-jvm.dmg** file.
- You will see the following screen on successful installation.



Figure 14: Weka Selection

Click on the **weka-3-8-3-corretto-JVM** icon to start Weka.

Optionally you may start it from the command line –

```
java -jar weka.jar
```

The WEKA GUI Chooser application will start and you would see the following screen –



Figure 15: Weka Home Screen

The GUI Chooser application allows you to run five different types of applications as listed here –

- Explorer
- Experimenter
- KnowledgeFlow
- Workbench
- Simple CLI



Install Weka and explore all the options of Weka Home Screen

WEKA Download And Installation

1) Download the software from https://waikato.github.io/weka-wiki/downloading_weka/

Check the configuration of the computer system and download the stable version of WEKA.

2) After a successful download, open the file location and double click on the downloaded file. The Step Up wizard will appear. Click on Next.



Figure 16: Set-up Wizard

3) The License Agreement terms will open. Read it thoroughly and click on "I Agree".

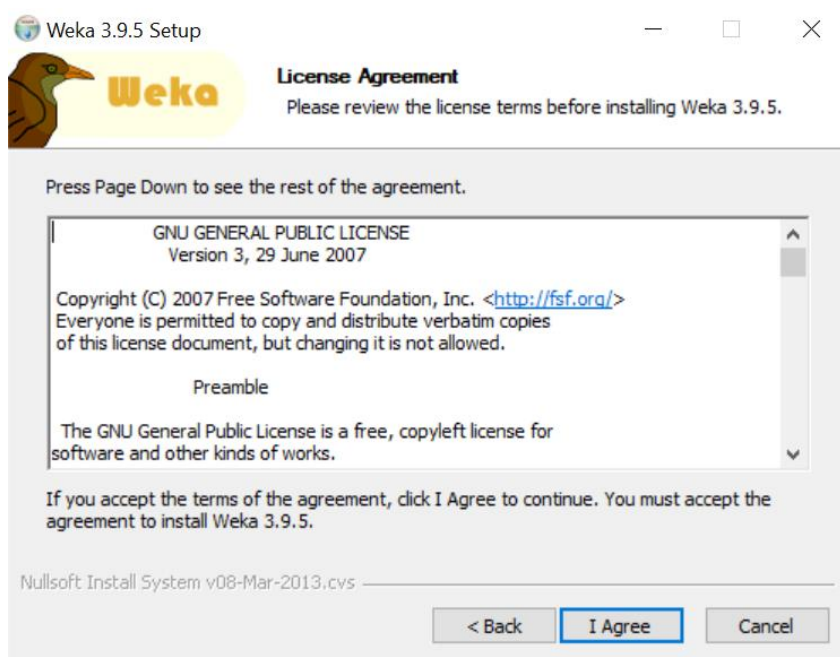


Figure17:

- 4) According to your requirements, select the components to be installed. Full component installation is recommended. Click on Next.

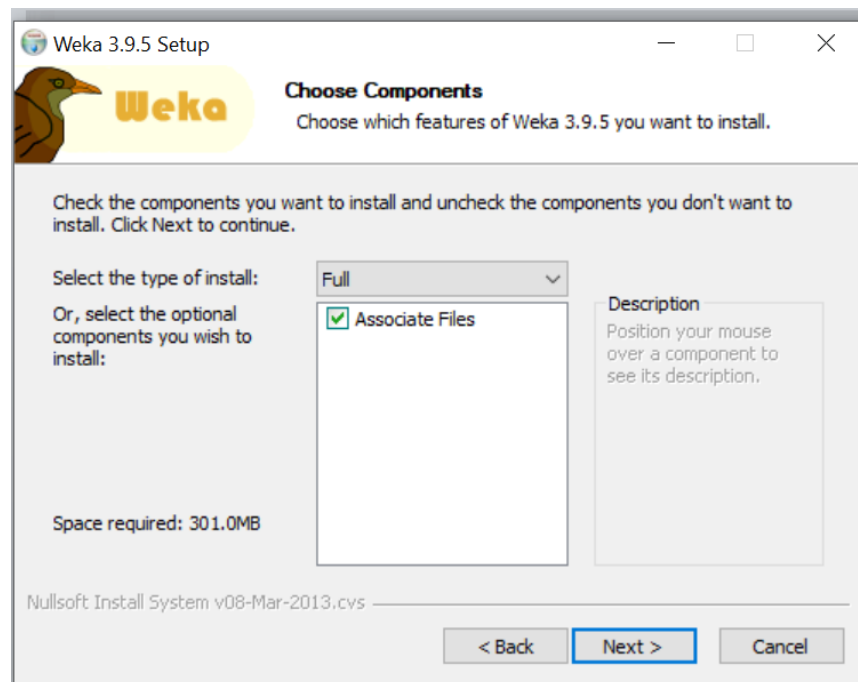


Figure18:

- 5) Select the destination folder and Click on Next.

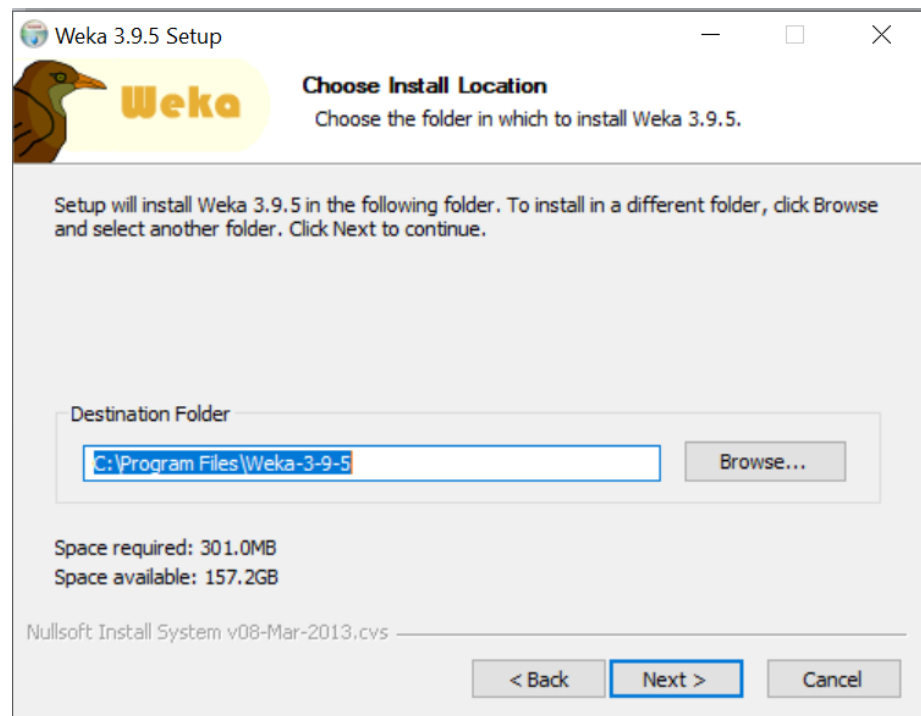


Figure19:

6) Then, Installation will start.

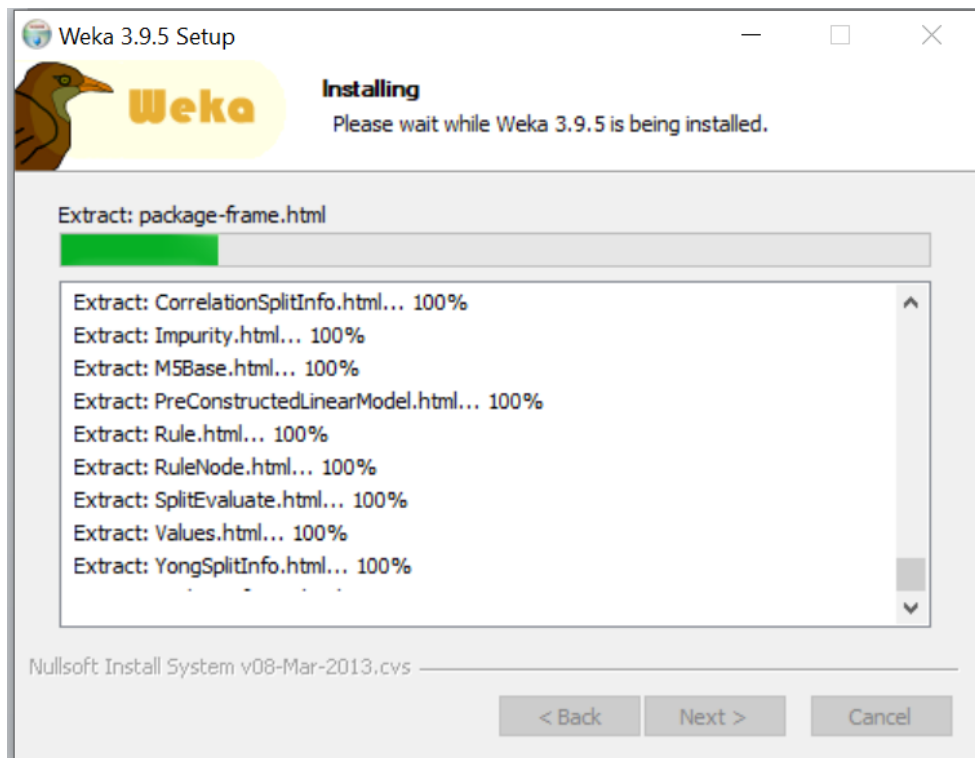


Figure20:

7) After the installation is complete, the following window will appear. Click on Next.

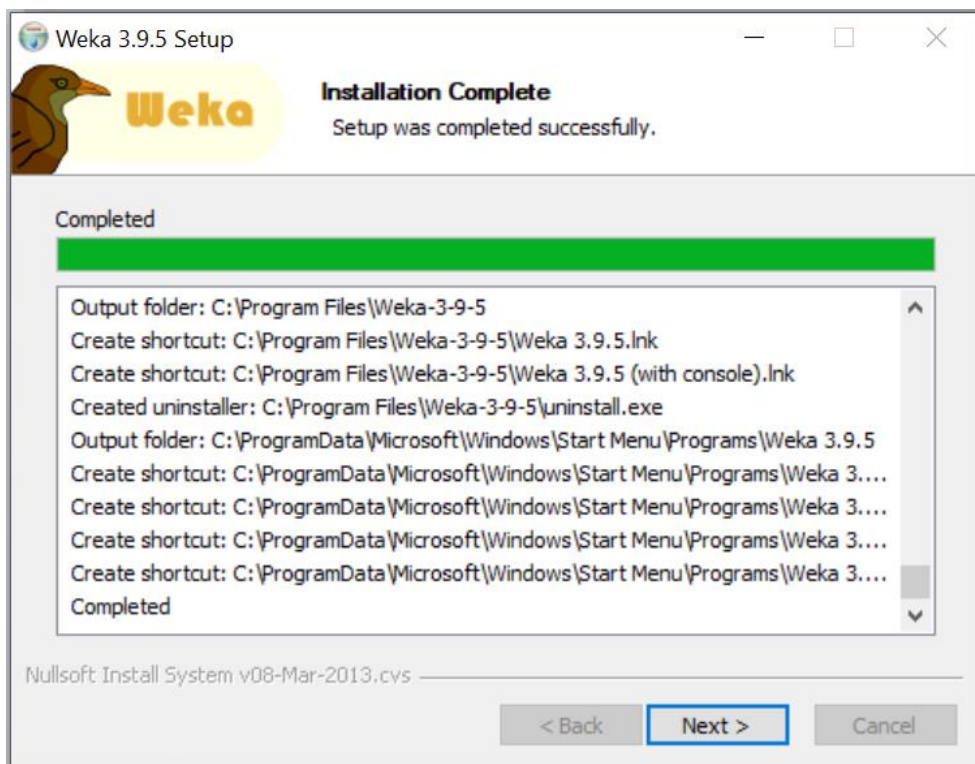


Figure21:

8) Select the Start Weka checkbox. Click on Finish.

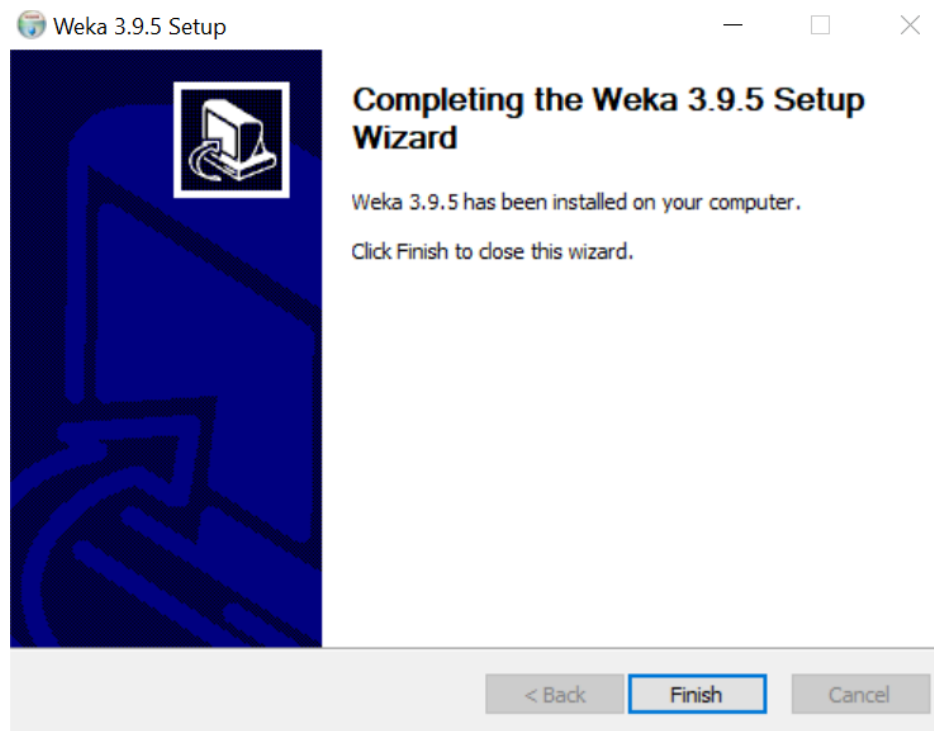


Figure22:

9) WEKA Tool and Explorer window opens.

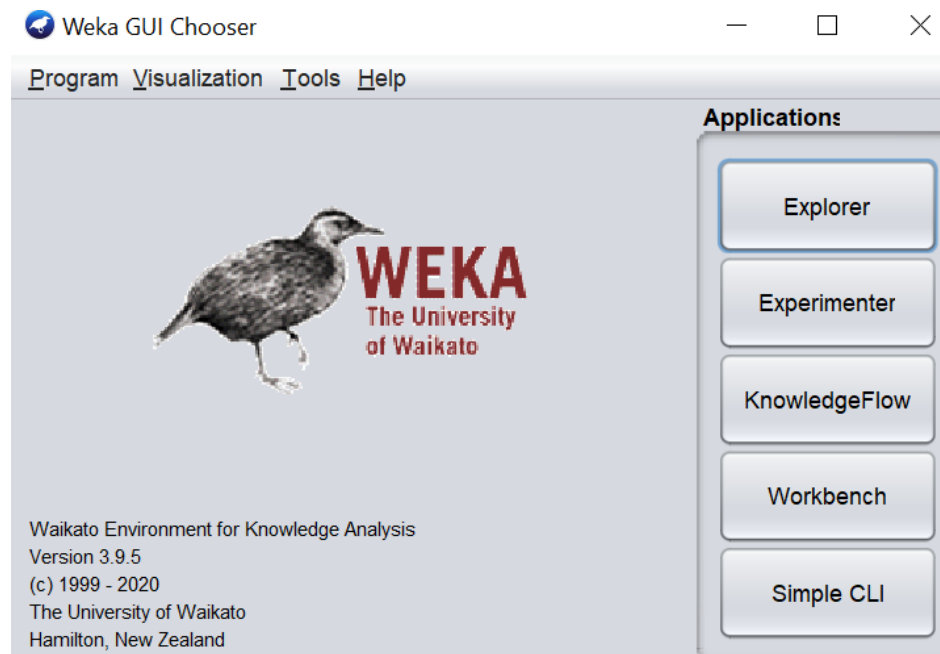


Figure23:



Install the current version of Weka on your system

Summary

- RapidMiner is a free-of-charge, open-source software tool for data and text mining.
- In addition to Windows operating systems, RapidMiner also supports Macintosh, Linux, and Unix systems.
- RapidMiner Studio is a visual data science workflow designer accelerating the prototyping & validation of models.
- With RapidMiner Studio, you can access, load, and analyze any type of data – both traditional structured data and unstructured data.
- Weka is a collection of machine learning algorithms for data mining tasks.
- Weka algorithms can either be applied directly to a dataset or called from your own Java code.
- Weka contains tools for data pre-processing, classification, regression, clustering, association rules, and visualization.

Keywords

Numeric: All kinds of number values; include the date, time, integer, and real numbers.

Text: Nominal data type that allows for more granular distinction (to differentiate from polynomial).

Delete Repository Entry: An operator to delete a repository entry within a process.

Specificity: Specificity relates to the classifier's ability to identify negative results.

Arff Format: An Arff file contains two sections - header and data.

Self Assessment

1. A database where all of the values for a particular column are stored contiguously is called?
 - a. Column-oriented storage
 - b. In-memory database
 - c. Partitioning
 - d. Data Compression
2. Data mining can also be applied to other forms such as
 - i) Data streams
 - ii) Sequence data
 - iii) Networked data
 - iv) Text data
 - v) Spatial data

Select the correct one:

- a. i, ii, iii, and v only
 - b. ii, iii, iv, and v only
 - c. I, iii, iv, and v only
 - d. All i, ii, iii, iv and v
3. The _____ is a symbolic representation of facts or ideas from which information can potentially be extracted.
 - a. knowledge.
 - b. data.
 - c. algorithm.
 - d. program.

4. Data mining is used to refer to the _____ stage in knowledge discovery in the database.
 - a. selection.
 - b. retrieving.
 - c. discovery.
 - d. coding.
5. Which of the following can be considered as the correct process of Data Mining?
 - a. Infrastructure, Exploration, Analysis, Interpretation, Exploitation
 - b. Exploration, Infrastructure, Analysis, Interpretation, Exploitation
 - c. Exploration, Infrastructure, Interpretation, Analysis, Exploitation
 - d. Exploration, Infrastructure, Analysis, Exploitation, Interpretation
6. Which of the following is an essential process in which the intelligent methods are applied to extract data patterns?
 - a. Warehousing
 - b. Data Mining
 - c. Text Mining
 - d. Data Selection
7. In data mining, how many categories of functions are included?
 - a. 5
 - b. 4
 - c. 2
 - d. 3
8. The issues like efficiency, scalability of data mining algorithms comes under _____
 - a. Performance issues
 - b. Diverse data type issues
 - c. Mining methodology and user interaction
 - d. All of the above
9. Which of the following statements about the query tools is correct?
 - a. Tools developed to query the database
 - b. Attributes of a database table that can take only numerical values
 - c. Both and B
 - d. None of the above
10. Which one of the following refers to the binary attribute?
 - a. This takes only two values. In general, these values will be 0 and 1, and they can be coded as one bit
 - b. The natural environment of a certain species
 - c. Systems that can be used without knowledge of internal operations
 - d. All of the above
11. Which of the following is the data mining tool?
 - a. Borland C.
 - b. Weka.
 - c. Borland C++.
 - d. Visual C.

12. Which one of the following issues must be considered before investing in data mining?
- Compatibility
 - Functionality
 - Vendor consideration
 - All of the above

Review Questions

- What is Rapid Miner? explain the various facilities provided by Rapid Miner.
- Explain the process of creating a user account in Rapid Miner.
- Elaborate on various Rapid Miner products.
- Write down the installation steps of Rapid Miner.
- How to install Weka in a windows environment. Write all the steps required for the installation.

Answers:

- | | | | |
|----|---|----|---|
| 1 | a | 2 | d |
| 3 | b | 4 | c |
| 5 | a | 6 | b |
| 7 | c | 8 | a |
| 9 | a | 10 | a |
| 11 | b | 12 | d |

Further Readings



Hofmann, M., & Klinkenberg, R. (Eds.). (2016). *RapidMiner: Data mining use cases and business analytics applications*. CRC Press.

Kotu, V., & Deshpande, B. (2014). *Predictive analytics and data mining: concepts and practice with rapidminer*. Morgan Kaufmann.

Witten, I. H., Frank, E., Trigg, L. E., Hall, M. A., Holmes, G., & Cunningham, S. J. (1999). *Weka: Practical machine learning tools and techniques with Java implementations*.

Garner, S. R. (1995, April). Weka: The Waikato environment for knowledge analysis. In *Proceedings of the New Zealand computer science research students conference* (Vol. 1995, pp. 57-64).



<https://rapidminer.com/get-started/>

<https://rapidminer.software.informer.com/5.3/>

https://waikato.github.io/weka-wiki/downloading_weka/

https://www.tutorialspoint.com/weka/weka_installation.htm